THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 4: **Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student heated a blue crystalline solid. It turned white and drops of liquid appeared on the cooler part of the tube. On adding water to the white powder, it turned blue and became warm. This indicates: 1

(a) a reversible chemical change **(b)** burning
(c) a physical change **(d)** an irreversible change

2. Which of these is the most reliable test to decide whether a change is chemical? 1

(a) change in shape **(b)** change in position
(c) formation of a new substance **(d)** change in size

3. A shop owner stores iron nails in a damp room. After some days, nails are covered with flaky reddish layer. The best advice to prevent this: 1

(a) store in water **(b)** store in dry place or oil them
(c) store in salt water **(d)** store in vinegar

4. An iron nail kept in blue solution X gets reddish-brown coating and the solution turns green. X is likely: 1

(a) copper sulphate **(b)** sugar **(c)** sodium chloride **(d)** lime water

5. Which of the following would you use to show that a new gas is formed? 1

(a) magnet **(b)** balance **(c)** ruler **(d)** lime water test

6. A farmer wants to keep tools from rusting during the rainy season. Which is best? 1

(a) leave outside **(b)** wash daily with salt water
(c) apply oil and keep dry **(d)** keep in a damp shed

7. A solution turns milky when a gas is passed. The gas is probably: 1

(a) hydrogen **(b)** oxygen **(c)** carbon dioxide **(d)** nitrogen

8. A student heats a piece of wax and sees it melt. On stronger heating, black smoke appears. The black smoke indicates: 1

(a) no change **(b)** melting only
(c) evaporation only **(d)** unburnt carbon, a chemical change

9. Why are bicycle chains regularly oiled? 1

(a) to make them brittle **(b)** to prevent rusting and reduce friction
(c) to make them heavier **(d)** to make them shiny only

10. An experiment shows that the mass of a burnt candle's products (including gases) equals the mass of the candle and oxygen used. This supports: 1

(a) melting **(b)** conservation of mass

(c) floating **(d)** rusting

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A student weighs an iron nail before and after rusting and finds an increase in mass. Explain. 2

12. Explain why the burning of paper is a chemical change but its tearing is a physical change. 2

13. Why is a slow chemical change not always noticed? 2

14. Explain why the melting of ice in a glass of water is physical but the burning of magnesium is chemical. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A student kept three iron nails: A in a closed jar with dry air and calcium chloride, B in boiled water with oil layer, C in a open beaker with tap water. Predict and justify results. 3

16. Explain why a mixture of salt and sand is physically separated but a mixture of iron and sulphur on heating is not. 3

17. Why is a firework display a mix of changes? 3

OR

Calcium oxide reacts with water and the mixture becomes hot. Is it physical or chemical? Give two reasons. 3

18. Explain with an example how a physical change may accompany a chemical change. 3

OR

Explain how oxygen takes part in rusting and burning. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Design an experiment to find out how three factors (air, water and salt) affect rusting, using six nails; describe the set-up, observations and conclusions. 5

OR

Explain how to conduct an experiment to show that burning needs oxygen and produces carbon dioxide. 5

20. Analyse the changes taking place when you light a gas stove under a vessel of milk, with milk boiling over and burning at the bottom. Classify each change. 5

OR

Discuss the use of the idea of physical and chemical change in cleaning methods like washing, dry cleaning, polishing silver and removing rust. 5

END OF QUESTION PAPER Total: 40 Marks